

Solving Equations & Inequalities

One- & Two-Step · Multi-Step · Fractions & Distribution · Rearranging Formulas · Inequalities

MTH1W · Class — Guided Practice

Name: _____

Date: _____

Goal: Solve one-step, two-step and multi-step equations, rearrange formulas, and solve and graph linear inequalities.

Warm-up (quick recall)

1. Solve: $x + 5 = 12$.

2. Solve: $3x = 18$.

3. Solve: $x - 4 = 10$.

Examples & Practice

For each skill, study the worked **Example**, then solve the questions that follow. Difficulty rises within each set (L1 → L3). Answers are on the last page.

One-Step Equations

Example:

Solve $x - 7 = 3$.

Add 7 to both sides: $x - 7 + 7 = 3 + 7$.

$x = 10$. Answer: $x = 10$.

4. Solve: $x + 9 = 15$. **[L1]**

5. Solve: $4x = 28$. **[L1]**

6. Solve: $x/5 = 6$. **[L2]**

7. Solve $-3x = 21$, then check your answer by substitution. **[L3]**

Two-Step Equations

Example:

Solve $2x + 3 = 11$.

Subtract 3: $2x = 8$. Divide by 2: $x = 4$.

Answer: $x = 4$.

8. Solve: $3x + 4 = 19$. **[L1]**

9. Solve: $x/3 - 5 = 1$. **[L2]**

10. Solve: $7 - 2x = 15$. **[L2]**

11. Solve $4x + 9 = 1$, then check by substitution. **[L3]**

Multi-Step Equations (variables on both sides)

Example:

Solve $5x - 2 = 3x + 8$.

Subtract $3x$: $2x - 2 = 8$. Add 2: $2x = 10$.

Divide by 2: $x = 5$.

12. Solve: $6x = 2x + 12$. **[L1]**

13. Solve: $8x - 5 = 3x + 15$. **[L2]**

14. Solve: $10 - x = 2x + 1$. **[L2]**

15. Solve $5x + 3 = 2x + 18$, then check by substitution. **[L3]**

Equations with Fractions and Distribution

Example:

Solve $2(x + 3) = 14$.

Distribute: $2x + 6 = 14$. Subtract 6: $2x = 8$.

Divide by 2: $x = 4$.

16. Solve: $3(x - 1) = 12$. **[L1]**

17. Solve: $(x + 2)/4 = 3$. **[L2]**

18. Solve: $5(2x - 1) = 25$. **[L2]**

19. Solve $4(x + 1) = 2(x + 7)$, then check by substitution. **[L3]**

Rearranging Formulas (solve for a variable)

Example:

Solve $y = mx + b$ for x .

Subtract b : $y - b = mx$. Divide by m .

$x = (y - b)/m$.

20. Solve $C = 2\pi r$ for r . **[L1]**

21. Solve $A = (1/2)bh$ for h . **[L2]**

22. Solve $P = 2l + 2w$ for w . **[L2]**

23. Solve $A = (1/2)(a + b)h$ for b . **[L3]**

Solving and Graphing Linear Inequalities

Example:

Solve and graph $x + 4 < 9$.

Subtract 4: $x < 5$.

Graph: open circle at 5, arrow pointing left.

24. Solve and describe the graph: $x - 3 > 2$. **[L1]**

25. Solve and describe the graph: $2x + 1 \leq 9$. **[L2]**

26. Solve and describe the graph: $3x \geq -6$. **[L2]**

27. Solve and describe the graph of $-2x < 8$. Explain what happens to the inequality sign. **[L3]**

Answers

Warm-up: W1) $x = 7$ W2) $x = 6$ W3) $x = 14$

- 4.** (L1) $x = 6$
- 5.** (L1) $x = 7$
- 6.** (L2) $x = 30$
- 7.** (L3) $x = -7$; check: $-3(-7) = 21$.
- 8.** (L1) $3x = 15$, so $x = 5$
- 9.** (L2) $x/3 = 6$, so $x = 18$
- 10.** (L2) $-2x = 8$, so $x = -4$
- 11.** (L3) $4x = -8$, $x = -2$; check: $4(-2)+9 = 1$.
- 12.** (L1) $4x = 12$, so $x = 3$
- 13.** (L2) $5x = 20$, so $x = 4$
- 14.** (L2) $9 = 3x$, so $x = 3$
- 15.** (L3) $3x = 15$, $x = 5$; check: $5(5)+3 = 28 = 2(5)+18$.
- 16.** (L1) $3x = 15$, so $x = 5$
- 17.** (L2) $x + 2 = 12$, so $x = 10$
- 18.** (L2) $10x - 5 = 25$, $10x = 30$, $x = 3$
- 19.** (L3) $4x+4 = 2x+14$, $2x = 10$, $x = 5$; check: both sides = 24.
- 20.** (L1) $r = C/(2\pi)$
- 21.** (L2) $2A = bh$, so $h = 2A/b$
- 22.** (L2) $2w = P - 2l$, so $w = (P - 2l)/2$
- 23.** (L3) $2A = (a+b)h$, $a+b = 2A/h$, $b = (2A - ah)/h$.
- 24.** (L1) $x > 5$; open circle at 5, arrow right.
- 25.** (L2) $2x \leq 8$, $x \leq 4$; closed circle at 4, arrow left.
- 26.** (L2) $x \geq -2$; closed circle at -2, arrow right.
- 27.** (L3) Divide by -2 and flip: $x > -4$; open circle at -4, arrow right.